Problem Set 2

Eigenvalues, symmetric and positive definite matrices, and the heat equation

18.085 Computational Science and Engineering I • Fall 2026 • G. G. Rozenman

Due Friday, October 23, 2026, at 11:59 pm ET

Covers Lectures 6–9 (Strang §1.5–1.6, with parts of §2.1 and §6.5). Seven problems, 175 points (25, 20, 15, 25, 15, 40, 35). This set is longer than PS1. Problems 1, 3 and 4 need only Lectures 6–7, so they can be started now.

Collaboration. You are encouraged to work together, but all final write-ups must be your own. List your discussion partners and any outside resources you consulted, including AI tools. Using an AI system in place of your own reasoning without saying so violates the collaboration policy; using one as a study aid and saying so does not.

Computing. Problem 7 is computational. Use Python or MATLAB, your choice, starting from `pset2_starter.py` or `pset2_starter.m` on Canvas. Attach your code and your plots to the PDF. Elsewhere a computer is welcome for checking arithmetic, but the hand computations asked for are the point.

Notation follows Lecture 7: $\boldsymbol{A} = \boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}^{-1}$, and $\boldsymbol{A} = \boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}^{\mathsf{T}}$ with $\boldsymbol{V}^{\mathsf{T}}\boldsymbol{V} = \boldsymbol{I}$ when $\boldsymbol{A}$ is symmetric. For a complex vector, $\boldsymbol{v}^{\dagger} = (\boldsymbol{v}^{*})^{\mathsf{T}}$.

1. Eigendecomposition, and a matrix that has none (25 pts) ★★⋆

From: L6–L7 · Strang 1.5

(a) Find the eigendecompositions $\boldsymbol{A} = \boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}^{-1}$ of

$$\boldsymbol{A}_1 = \begin{bmatrix} 2 & 3 \\ 0 & 5 \end{bmatrix}, \qquad \boldsymbol{A}_2 = \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}.$$

For each, compute $\boldsymbol{A}^2$ directly and check that it equals $\boldsymbol{V}\boldsymbol{\Lambda}^2\boldsymbol{V}^{-1}$. $\boldsymbol{A}_2$ has a special form; write it as a column times a row and use that to explain the value of $\boldsymbol{A}_2^2$ in one line.

(b) Attempt the same for

$$\boldsymbol{B} = \begin{bmatrix} 1 & 4 \\ -1 & 5 \end{bmatrix}.$$

Find all eigenvalues and all eigenvectors of $\boldsymbol{B}$, and explain exactly where the construction of $\boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}^{-1}$ breaks down.

(c) Powers and the exponential of $\boldsymbol{B}$ anyway. Write $\boldsymbol{B} = 3\boldsymbol{I} + \boldsymbol{N}$. Compute $\boldsymbol{N}^2$, and use it to show that

$$\boldsymbol{B}^k = 3^k\boldsymbol{I} + k\,3^{k-1}\boldsymbol{N} \qquad (k \geq 1)$$

(why is the binomial theorem allowed here?). Sum the series for $e^{\boldsymbol{B}t}$ to get $e^{\boldsymbol{B}t} = e^{3t}(\boldsymbol{I} + t\boldsymbol{N})$, and solve $d\boldsymbol{x}/dt = \boldsymbol{B}\boldsymbol{x}$ with $\boldsymbol{x}(0) = (1, 0)$. Which feature of your answer could never come out of a formula $\boldsymbol{V}e^{\boldsymbol{\Lambda}t}\boldsymbol{V}^{-1}$?

2. A rank-one matrix (20 pts) ★★☆

From: L6–L8 · Strang 1.5–1.6

Consider the symmetric 4×4 matrix

$$\boldsymbol{M} = \begin{bmatrix} 1 & 2 & 2 & 4 \\ 2 & 4 & 4 & 8 \\ 2 & 4 & 4 & 8 \\ 4 & 8 & 8 & 16 \end{bmatrix}.$$

(a) Find $\boldsymbol{M} = \boldsymbol{L}\boldsymbol{U}$ by hand. How many nonzero pivots are there? Show that the product $\boldsymbol{L}\boldsymbol{U}$ collapses to a single column times a single row, and write $\boldsymbol{M} = \boldsymbol{u}\boldsymbol{u}^{\mathsf{T}}$.

(b) Compute $\boldsymbol{M}\boldsymbol{u}$ and $\boldsymbol{M}\boldsymbol{x}$ for any $\boldsymbol{x} \perp \boldsymbol{u}$, and read off all four eigenvalues. Find three mutually orthogonal eigenvectors for $\lambda = 0$ with integer entries, and write $\boldsymbol{M} = \boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}^{\mathsf{T}}$ with $\boldsymbol{V}^{\mathsf{T}}\boldsymbol{V} = \boldsymbol{I}$. Check your eigenvalues against the trace.

(c) Shifting by the identity. For a real number c, find the eigenvalues of $\boldsymbol{I} + c\,\boldsymbol{M}$. For which c is $\boldsymbol{I} + c\,\boldsymbol{M}$ (i) singular, (ii) positive definite, (iii) orthogonal? For c not in case (i), verify that

$$(\boldsymbol{I} + c\,\boldsymbol{u}\boldsymbol{u}^{\mathsf{T}})^{-1} = \boldsymbol{I} - \frac{c}{1 + c\,\boldsymbol{u}^{\mathsf{T}}\boldsymbol{u}}\,\boldsymbol{u}\boldsymbol{u}^{\mathsf{T}}.$$

Describe geometrically what the orthogonal matrix with $c \neq 0$ does to $\boldsymbol{u}$ and to vectors perpendicular to $\boldsymbol{u}$.

3. Orthogonal matrices (15 pts) ★★☆

From: L4, L7 · Strang 1.5

A real $n \times n$ matrix $\boldsymbol{Q}$ is orthogonal if $\boldsymbol{Q}^{\mathsf{T}}\boldsymbol{Q} = \boldsymbol{I}$.

(a) Show that $\|\boldsymbol{Q}\boldsymbol{x}\| = \|\boldsymbol{x}\|$ for every $\boldsymbol{x} \in \mathbb{R}^n$. The columns $\boldsymbol{q}_1, \ldots, \boldsymbol{q}_n$ of $\boldsymbol{Q}$ are an orthonormal basis; what are the coefficients of $\boldsymbol{x}$ in that basis, and why does this make the equality of lengths unsurprising?

(b) Show that if $\boldsymbol{Q}_1$ and $\boldsymbol{Q}_2$ are orthogonal, so is $\boldsymbol{Q}_1\boldsymbol{Q}_2$. Is $\boldsymbol{Q}_1 + \boldsymbol{Q}_2$ always orthogonal?

(c) Eigenvalues on the unit circle. Suppose $\boldsymbol{Q}\boldsymbol{v} = \lambda\boldsymbol{v}$, where λ and $\boldsymbol{v} \neq \boldsymbol{0}$ may be complex. Compute $(\boldsymbol{Q}\boldsymbol{v})^{\dagger}(\boldsymbol{Q}\boldsymbol{v})$ in two ways to show that $|\lambda| = 1$. Then find the eigenvalues and eigenvectors of the rotation

$$\boldsymbol{R}_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}.$$

(d) Antisymmetric generators. Suppose $\boldsymbol{S}^{\mathsf{T}} = -\boldsymbol{S}$. Show that $e^{\boldsymbol{S}t}$ is orthogonal for every t. (You may use $(e^{\boldsymbol{X}})^{\mathsf{T}} = e^{\boldsymbol{X}^{\mathsf{T}}}$ and $e^{\boldsymbol{X}}e^{\boldsymbol{Y}} = e^{\boldsymbol{X}+\boldsymbol{Y}}$ when $\boldsymbol{X}\boldsymbol{Y} = \boldsymbol{Y}\boldsymbol{X}$.) For

$$\boldsymbol{S} = \begin{bmatrix} 0 & -\omega \\ \omega & 0 \end{bmatrix}, \qquad \omega \neq 0,$$

compute $\boldsymbol{S}^2$ and sum the series to show $e^{\boldsymbol{S}t} = \boldsymbol{R}_{\omega t}$. What does this say about the solutions of $dx_1/dt = -\omega x_2$, $dx_2/dt = \omega x_1$?

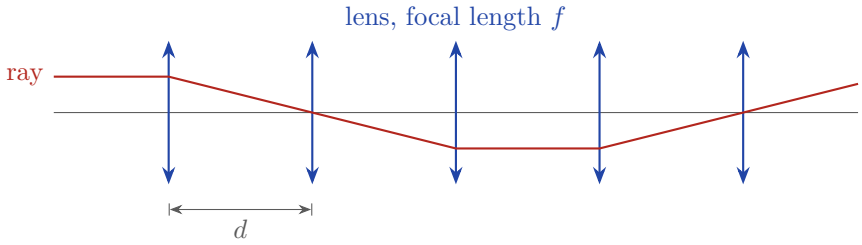
4. Lens waveguides and powers of a matrix (25 pts) ★★★

From: L3 (ray matrices), L6–L7 · new problem

A paraxial ray is described by $\boldsymbol{r} = (y, \theta)$: its height y above the optical axis (in mm) and its angle θ to the axis (in radians). Lecture 3 used two 2×2 matrices:

$$\text{free space of length } d: \quad \boldsymbol{P} = \begin{bmatrix} 1 & d \\ 0 & 1 \end{bmatrix}, \qquad \text{thin lens of focal length } f: \quad \boldsymbol{L} = \begin{bmatrix} 1 & 0 \\ -1/f & 1 \end{bmatrix}.$$

A lens waveguide is a long row of identical thin lenses spaced d apart. One period (travel d, then pass a lens) acts by $\boldsymbol{M} = \boldsymbol{L}\boldsymbol{P}$, so after N periods $\boldsymbol{r}_N = \boldsymbol{M}^N \boldsymbol{r}_0$.



(a) Show that

$$\boldsymbol{M} = \begin{bmatrix} 1 & d \\ -1/f & 1 - d/f \end{bmatrix},$$

that $\det \boldsymbol{M} = 1$, and that the eigenvalues satisfy $\lambda^2 - \tau\lambda + 1 = 0$ with $\tau = 2 - d/f$. Show that:

- if $|\tau| < 2$, the eigenvalues are $e^{\pm i\varphi}$ with $\cos\varphi = \tau/2$;
- if $|\tau| > 2$, the eigenvalues are real, with product 1, and one has $|\lambda| > 1$.

In each case the two eigenvalues are distinct, so $\boldsymbol{M}$ has two independent eigenvectors. Use $\boldsymbol{r}_0 = c_1\boldsymbol{v}_1 + c_2\boldsymbol{v}_2$ to explain why every ray stays bounded in the first case, and why almost every ray escapes in the second. Conclude that, for $d > 0$, the waveguide guides every ray when $d < 4f$ and not when $d > 4f$. (Part (d) settles $d = 4f$.)

(b) Take $f = 50\,\text{mm}$ and $d = 100\,\text{mm}$. Find the eigenvalues, and use $\boldsymbol{\Lambda}^2$ to show that $\boldsymbol{M}^2 = -\boldsymbol{I}$ without multiplying out $\boldsymbol{M}^2$. Then check by multiplying. What happens to every ray after two periods, and after four?

(c) Take $f = 50\,\text{mm}$ and $d = 250\,\text{mm}$. Find the eigenvalues and eigenvectors exactly and to four decimals. Expand $\boldsymbol{r}_0 = (1, 0)$ (a ray entering 1 mm above the axis, parallel to it) in the eigenvectors, and give a formula for its height y_N. Evaluate y_{10}, and check it by computing $\boldsymbol{M}^{10}\boldsymbol{r}_0$ directly (a computer is fine here). By what factor does the height grow per period for large N?

(d) The edge of stability. Take $f = 50\,\text{mm}$ and $d = 4f = 200\,\text{mm}$. Show that $\lambda = -1$ is a double eigenvalue and that $\boldsymbol{M}$ is defective. With $\boldsymbol{E} = \boldsymbol{M} + \boldsymbol{I}$, show $\boldsymbol{E}^2 = \boldsymbol{0}$ and $\boldsymbol{M}^k = (-1)^k(\boldsymbol{I} - k\boldsymbol{E})$. Find y_k for $\boldsymbol{r}_0 = (1, 0)$. Every eigenvalue has $|\lambda| = 1$; does the ray stay bounded? Which rays do?

5. The matrix A^TCA (15 pts) ★★★

From: L7–L8 · Strang 1.6, 2.1

Let A be $m \times n$ and C be symmetric $m \times m$.

(a) Show that A^TCA is symmetric. What is its shape?

(b) Show that A^TA has no negative entries on its diagonal. When is a diagonal entry zero?

(c) **Positive weights give positive definite.** Suppose $C = \text{diag}(c_1, \ldots, c_m)$ with every $c_k > 0$ and the columns of A are independent. Show that A^TCA is positive definite. Then take

$$A = \begin{bmatrix} 1 & 0 \\ -1 & 1 \\ 0 & -1 \end{bmatrix}, \qquad C = \text{diag}(2, 3, 6)$$

(two masses between three springs, both ends fixed; Au lists the spring stretches). Compute A^TCA, its pivots and its eigenvalues.

(d) Remove springs. (i) Delete the last spring (the last row of A and the last c_k). (ii) Keep only the middle spring. In each case compute A^TCA and decide whether it is positive definite or only semidefinite. If only semidefinite, find a nonzero u with $u^TA^TCAu = 0$ and say what it means physically.

6. Positive definite matrices and their square roots (40 pts) ★★☆

From: L7–L8 · Strang 1.5–1.6

(a) Decide whether each matrix is positive definite, positive semidefinite but not definite, or indefinite, using pivots. (The aside in Lecture 8 helps: with $M = L_0DL_0^T$, $x^TMx = \sum_i d_i z_i^2$ where $z = L_0^Tx$.) If a matrix is semidefinite but not definite, give a nonzero x with $x^TMx = 0$; if it is indefinite, give a nonzero x with $x^TMx < 0$.

$$A = \begin{bmatrix} 5 & -2 \\ -2 & 2 \end{bmatrix}, \qquad B = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix}, \qquad C = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 3 & 1 \\ 0 & 1 & 1 \end{bmatrix}.$$

(b) Find the eigenvalues and orthonormal eigenvectors of A, and compute e^{-At}. Write the solution of $dx/dt = -Ax$ for a general $x(0)$. For which $x(0)$ does the solution decay like e^{-6t}? In which direction does $x(t)$ point for large t, starting from a generic $x(0)$?

(c) Given a diagonal Λ with every $\Lambda_{jj} \geq 0$, write down a diagonal $\sqrt{\Lambda}$ with $\sqrt{\Lambda}\sqrt{\Lambda} = \Lambda$.

(d) For a symmetric positive semidefinite S, use $S = V\Lambda V^T$ and (c) to construct a symmetric positive semidefinite $\sqrt{S}$ with $\sqrt{S}\sqrt{S} = S$. Compute $\sqrt{A}$ exactly. Then show that A has exactly four square roots (matrices R with $R^2 = A$) and that only one of them is positive semidefinite. (Hint: R commutes with A. What does R do to an eigenvector of A?)

(e) Let L be an invertible lower triangular matrix. Prove that LL^T is positive definite.

(f) Let $\boldsymbol{S}$ be symmetric positive definite. Show that $\boldsymbol{S} = \boldsymbol{L}\boldsymbol{L}^{\mathsf{T}}$ for a lower triangular $\boldsymbol{L}$ (the *Cholesky factorization*). (Hint: start from $\boldsymbol{S} = \boldsymbol{L}_0\boldsymbol{D}\boldsymbol{L}_0^{\mathsf{T}}$ of Lecture 3, with $\boldsymbol{L}_0$ unit lower triangular.)

(g) Cholesky in practice. Compute the Cholesky factor $\boldsymbol{L}$ of $\boldsymbol{A}$ and of

$$\boldsymbol{K}_3 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}.$$

Carry out the same construction for $\boldsymbol{B}$ and for $\boldsymbol{C}$ by hand. What is different about the result for $\boldsymbol{B}$, and what goes wrong for $\boldsymbol{C}$? Explain why attempting a Cholesky factorization is a practical test for positive definiteness, and why a matrix like $\boldsymbol{B}$ is a hard case for it in floating point.

7. The heat equation by matrix exponentials (35 pts) ★★⋆

From: L7–L9 · Strang 1.5, 6.5

We solve the heat equation of Lecture 9 on a rod of length 1,

$$\frac{\partial u}{\partial t} = \gamma \frac{\partial^2 u}{\partial x^2}, \qquad 0 < x < 1, \qquad \gamma = 0.02,$$

starting from a hot section placed off centre,

$$u(x,0) = g(x) = \begin{cases} 1, & 0.1 < x < 0.4, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Fixed temperatures. Both ends are held at $u(0,t) = u(1,t) = 0$. Use the grid $x_j = jh$, $j = 1, \dots, n$, $h = 1/(n+1)$, and the centred second difference. Show that the vector $\boldsymbol{u}(t) = (u_1(t), \dots, u_n(t))$ of approximate temperatures satisfies $d\boldsymbol{u}/dt = \boldsymbol{A}\boldsymbol{u}$ with $\boldsymbol{A} = -(\gamma/h^2)\boldsymbol{K}_n$.

(b) Write the solution with a matrix exponential, and then with the eigendecomposition of $\boldsymbol{A}$. Show, using Lecture 8, that $\|\boldsymbol{u}(t)\|$ strictly decreases as long as $\boldsymbol{u} \neq \boldsymbol{0}$. (Differentiate $\frac{1}{2}\|\boldsymbol{u}\|^2$.)

(c) With $n = 100$, compute the eigendecomposition of $\boldsymbol{A}$. Plot the three eigenvectors whose eigenvalues are smallest in absolute value. Compare those three values of $|\lambda|$ with $\gamma\pi^2 k^2$, $k = 1, 2, 3$.

(d) Evaluate your solution from (b) at $t = 0$, 0.5, 2, 8, 30 and plot the five profiles on one figure. Repeat with n doubled and check that the curves barely change.

(e) Insulated ends. Now no heat crosses either end: $\partial u/\partial x = 0$ at $x = 0$ and $x = 1$. Use the cell-centred grid $x_j = (j - \frac{1}{2})h$, $j = 1, \dots, n$, $h = 1/n$, and the ghost values $u_0 = u_1$, $u_{n+1} = u_n$. Show that this gives $\boldsymbol{A} = -(\gamma/h^2)\boldsymbol{B}_n$, where $\boldsymbol{B}_n$ is $\boldsymbol{K}_n$ with its two corner entries changed to 1. Repeat (c) and (d) for this $\boldsymbol{A}$; in (c), compare with $\gamma\pi^2 k^2$ for $k = 0, 1, 2$.

(f) Where the heat goes. For the insulated rod: (i) show that $\boldsymbol{1}^{\mathsf{T}}\boldsymbol{B}_n = \boldsymbol{0}^{\mathsf{T}}$, and conclude that the total heat $Q(t) = h\sum_j u_j(t)$ is exactly constant. What is Q? (ii) Use the eigenvector for $\lambda = 0$ to find $\lim_{t\to\infty} \boldsymbol{u}(t)$. (iii) The approach to that limit is governed by the smallest nonzero $|\lambda|$. Estimate how long it takes the deviation from the limit to shrink by a factor of 100, and check against your computed profiles at $t = 8$ and $t = 30$.